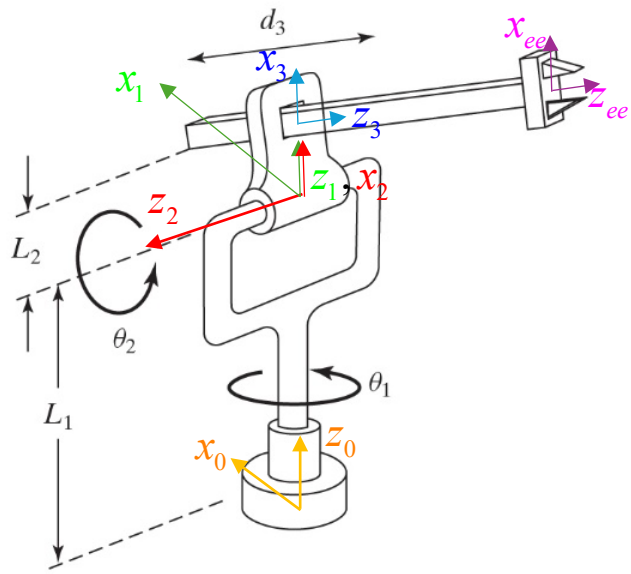
1-



0.5 for each frame origin and 0.75 for each frame axes

i-1	α_{i-1}	a_{i-1}	θ_i	d_i	i
0	0	0	θ_1	L_1	1
1	90	0	θ_2	0	2
2	90	L_2	0	d_3	3
3	0	L_3	0	0	ee

0.5 for zero elements and 0.75 for all nonzero and joint parameter elements.

$$
{}^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1} \\ \sin\theta_i\cos\alpha_{i-1} & \cos\theta_i\cos\alpha_{i-1} & -\sin\alpha_{i-1} & -d_i\sin\alpha_{i-1} \\ \sin\theta_i\sin\alpha_{i-1} & \cos\theta_i\sin\alpha_{i-1} & \cos\alpha_{i-1} & d_i\cos\alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}T_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{1}T_2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \sin\theta_1 & \cos\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{2}T_3 = \begin{bmatrix} 1 & 0 & 0 & L_2 \\ 0 & 0 & -1 & -d_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{3}T_{ee} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}T_{ee} = {}^{0}T_1{}^{1}T_2{}^{2}T_3{}^{3}T_{ee} = \begin{bmatrix} c_1c_2 & s_1 & c_1s_2 & c_1((L_2+L_3)c_2 + d_3s_2) \\ s_1c_2 & -c_1 & s_1s_2 & s_1((L_2+L_3)c_2 + d_3s_2) \\ s_2 & 0 & -c_2 & (L_2+L_3)s_2 - d_3c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

2 for each individual transformation matrix and 2 for the final answer

2-

$\alpha_1 = 90, a_1 = 0, d_1 = L_1, \alpha_2 = 90, a_2 = L_2, d_2 = 0, \theta_3 = 0, \alpha_3 = 0, a_3 = L_3, d_4 = 0$

Using Pieper's method, for $a_1 = 0$ 2

$r = k_3$ 2

and

$k_3 = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$

$$f_1 = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$
$$f_2 = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$
$$f_3 = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

Sub $\alpha_1 = 90, d_1 = L_1, \alpha_2 = 90, a_2 = L_2, d_2 = 0, \theta_3 = 0, \alpha_3 = 0, a_3 = L_3, d_4 = 0$

$$f_1 = L_2 + L_3$$
$$f_2 = -d_3$$
$$f_3 = 0$$
$$k_3 = (L_2 + L_3)^2 + d_3^2$$

2 for each *f* and 2 for k_3

We know

$$r = x^2 + y^2 + z^2$$
$$\rightarrow d_3 = \sqrt{x^2 + y^2 + z^2 - (L_2 + L_3)^2}$$

2 for d_3

We also have

$z = (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4$

and

$$k_1 = f_1 = L_2 + L_3$$
$$k_2 = -f_2 = d_3$$
$$k_4 = f_3 c\alpha_1 + d_2 c\alpha_1 = 0$$
$$\rightarrow z = (L_2 + L_3) s_2 - d_3 c_2$$

2 for each *k* and 2 for *z*

$u = \tan\frac{\theta_2}{2}$, $\cos\theta_2 = \frac{1-u^2}{1+u^2}$, $\sin\theta_2 = \frac{2u}{1+u^2}$ 4 for using the transcendental approach

$$\rightarrow z = (L_2 + L_3)\frac{2u}{1+u^2} - d_3\frac{1-u^2}{1+u^2}$$

$$(z-d_3)u^2 - 2(L_2+L_3)u + z + d_3 = 0$$

2 for forming the quadratic equation for u

$$u = \frac{L_2 + L_3 \pm \sqrt{(L_2+L_3)^2 - (z-d_3)(z+d_3)}}{(z-d_3)}$$

or

$$u = \frac{L_2 + L_3 \pm \sqrt{(L_2+L_3)^2 + d_3{}^2 - z^2}}{(z-d_3)}$$

2 for u

and

$$\theta_2 = 2\tan^{-1} u$$

2 for θ_2

Now for θ_1

$$c_1 g_1 - s_1 g_2 = x$$
$$s_1 g_1 + c_1 g_2 = y$$

1 for each θ_1 equation

and

$$g_1 = c_2 f_1 - s_2 f_2 + a_1 = c_2(L_2 + L_3) + s_2 d_3$$
$$g_2 = s_2 c\alpha_1 f_1 + c_2 c\alpha_1 f_2 - s\alpha_1 f_3 - d_2 s\alpha_1 = 0$$

2 for each g

$$\rightarrow$$
$$x = c_1(c_2(L_2 + L_3) + s_2 d_3)$$
$$y = s_1(c_2(L_2 + L_3) + s_2 d_3) \rightarrow \theta_1 = a\tan 2(y, x)$$

2 for θ_1

3-

a)

$$ {}^{3}R_6 = \begin{bmatrix} c_4 & -s_4 & 0 \\ s_4 & c_4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_5 & -s_5 & 0 \\ 0 & 0 & -1 \\ s_5 & c_5 & 0 \end{bmatrix} \begin{bmatrix} c_6 & -s_6 & 0 \\ 0 & 0 & 1 \\ -s_6 & -c_6 & 0 \end{bmatrix} $$

$$ = \begin{bmatrix} c_4c_5c_6 - s_4s_6 & -c_4c_5s_6 - s_4c_6 & -c_4s_5 \\ s_4c_5c_6 + c_4s_6 & -s_4c_5s_6 + c_4c_6 & -s_4s_5 \\ s_5c_6 & -s_5s_6 & c_5 \end{bmatrix} $$

5 for multiplying the three matrices

b)

$$ {}^{0}R_6 = \begin{bmatrix} 0 & -\mathrm{Cos}\,60 & \mathrm{Cos}\,30 \\ 0 & \sin 60 & \sin 30 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -0.5 & 0.866 \\ 0 & 0.866 & 0.5 \\ -1 & 0 & 0 \end{bmatrix} $$ 2.5

c)

$$ {}^{0}R_3 = \begin{bmatrix} c_1c_2 & s_1 & c_1s_2 \\ s_1c_2 & -c_1 & s_1s_2 \\ s_2 & 0 & -c_2 \end{bmatrix} $$

$$ {}^{0}R_3^{-1} = {}^{0}R_3^{T} = \begin{bmatrix} c_1c_2 & s_1c_2 & s_2 \\ s_1 & -c_1 & 0 \\ c_1s_2 & s_1s_2 & -c_2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0.707 \\ 0.707 & -0.707 & 0 \\ 0.5 & 0.5 & -0.707 \end{bmatrix} $$ 1.5

$$ {}^{3}R_6 = {}^{0}R_3^{-1}\,{}^{0}R_6 = \begin{bmatrix} 0.5 & 0.5 & 0.707 \\ 0.707 & -0.707 & 0 \\ 0.5 & 0.5 & -0.707 \end{bmatrix} \begin{bmatrix} 0 & -0.5 & 0.866 \\ 0 & 0.866 & 0.5 \\ -1 & 0 & 0 \end{bmatrix} $$

$$ = \begin{bmatrix} -0.707 & 0.183 & 0.683 \\ 0 & -0.966 & 0.259 \\ 0.707 & 0.183 & 0.683 \end{bmatrix} == \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

1 for finding the elements of ${}^{3}R_6$

$$ \sin\theta_5 = \pm\sqrt{r_{13}^{\ 2} + r_{23}^{\ 2}} $$ 1 for finding $\sin\theta_5$

$$\theta_5 = \begin{cases} a\tan 2(\sqrt{r_{13}{}^2 + r_{23}{}^2}, r_{33}) = 47 \\ a\tan 2(-\sqrt{r_{13}{}^2 + r_{23}{}^2}, r_{33}) = -47 \end{cases}$$

4 for each θ_5

$$\theta_6 = \begin{cases} a\tan 2(-r_{32}, r_{31}) = -14.53 \\ a\tan 2(r_{32}, -r_{31}) = 165.5 \end{cases}$$

4 for each θ_6

$$\theta_4 = \begin{cases} a\tan 2(-r_{23}, -r_{13}) = 200.8 \\ a\tan 2(r_{23}, r_{31}) = 20.8 \end{cases}$$

4 for each θ_4